The gauge theorem in type B: one collapse law for every root of B_N, and a domain cut out by the root system itself

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Abstract

Let $\psi_X(v) = (1-qv^h)/(1-q)$ with $h = 2$ for $X = C$ and $h = 1$ for $X = B$, $\varphi_X = \psi_X^{-1}$, and $\widehat{A}^X_{\mathbf{M},\mathbf{I}} := \widetilde{A}^X_{\mathbf{M},\mathbf{I}} \prod_{m\in\mathbf{M}} \psi_X(w_m)$, where $\widetilde{A}^X = W^X/\mathcal{E}^X$ is Motegi's normalised type-X weight function on the 2^N fixed points of $LG(N, 2N)$. This paper closes gap **G2** of `2026-10-18-c2`: for every positive root α of B_N, every N, every $\mathbf{M}$ in the colliding stratum and every $\mathbf{I}$, the collapse multiplier is $\rho^\alpha = \prod_{m\in D(\alpha)} \psi_B(w_m)^{-1}$; hence $\widehat{A}^B$ collapses with multiplier exactly 1 at every root of B_N. The pre-registration was right: the ι-orbit classification is type-independent, only h moves, and the multiplier of $\widehat{A}$ is again 1. Measured 104/104 at all nine positive roots of B_2 and B_3 by two mechanisms, with 124/124 vacuous strata accounted separately and three admissible plants firing 13/13 each.

The proof is one computation, and it puts h in two places rather than one. **(1)** Cancelling W^B's prefactor $\prod_k (1+u_k^{-1})^{-1}$ against line 3c's (u_j+1) — which must be done *analytically*, since the first has a pole exactly where the second has a zero — turns the type-B summand into the type-C summand times $\prod_{\text{heads}} v_j/(v_j+1)$. Equivalently: **the type-B head factor is (v_j-1) where type C's is (v_j^2-1)** — that is $1-e^{-\alpha}$ for the boundary root, short in B and long in C. Since the head factor is the same on both sides of an adjunction ratio, the *unnormalised* ratio is type-independent and Steps 1–3 are not analogous to type C's but identical to them. **(2)** The whole remaining difference is the u-diagonal block of the normaliser, and there the surprise: the factor $\prod_y (1-qy^2)^{-1}$ produced by Lemma I2 — the lemma that in type C *is* where ψ comes from — is in type B cancelled exactly by $\mathcal{E}^B$'s own $\prod_j (1-qu_j^2)$, and ψ_B comes instead from $\mathcal{E}^B$'s numerator $\prod_j (1-qu_j)$. **The two ψ's have different origins**; they are not one formula with a parameter.

A pre-registered claim of mine was then **refuted, and the refutation is the best thing here.** I expected the arbitrary-ι-closed theorem to have content at the second τ-fixed point $\varepsilon = -1$, with $\varphi_B(-1) = (1-q)/(1+q)$ as a sharp type-discriminator. It does not: (v_j-1) vanishes only at $v_j = +1$, so **type-B head exclusion fails at -1** (measured: head term $\neq 0$, 9/9), and the law fails there — 0/9 at $r \geq 1$, against 6/6 at $r = 0$ where there is no head

slot at all. So the hypothesis is *necessary*, and what replaces my claim is better: the τ-fixed points at which type-X head exclusion fires are exactly the h_X-th roots of unity, i.e. exactly the points of the boundary root hyperplane $\{w_m^h = 1\}$. **The theorem's domain in each type is cut out by that type's own root system**, and reaches no further.

Two provenance corrections, both against me. **(i)** Task 0 of my brief ordered a literature check before proving. Motegi's `normalizedsusythree` (L2767–2771) is the type-B pair multiplier and `stabilityWB` (L2779–2783) the type-B boundary one; worse, `normalizedsusytwo` (L2618–2622) is the *type-C* pair multiplier — the headline number of `2026-10-18-c2`, whose priority section credited Rick and an earlier session of mine and did not cite Motegi, and which sits 27 lines above a statement my own engine docstring already cited. **(ii)** The vacuous strata $r > |s_\alpha \mathbf{M}|$ vanish by head exclusion, not by triangularity; `2026-10-18-c2`'s `cor:heads` cites (M3) there and that citation is wrong, though its conclusion is right.

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1 What this session was asked to do, and what happened

The brief named one theorem, G2, and put a mandatory half-hour of literature checking in front of it, because the previous session had spent part of itself proving a statement (G3, $\widetilde{A} \in R$) that was already in Motegi four lines from a citation I had been handed. The order was: read the relevant propositions *first*, and if they force type B, say so and grade G2 a corollary rather than a theorem. Outcome, in the order it happened:

1. **The disambiguation the brief demanded, resolved by the defining sentence rather than the phrase.** Two things in my notes are called "the type B gap". **Q514** asks whether Motegi's `normalizationWB` (L1386) is a product over the positive roots of B_N whose A-type factors are the same $(1 - w_a/w_{i_l})$: it is about the *diagonal* $\widetilde{A}_{\mathbf{I},\mathbf{I}}$ and the A-type-divisor colength, and it was closed on 10-18. **G2** is about ρ^α, the *collapse multiplier at a root hyperplane*, for all three families. Different objects; Q514 closed, G2 open.

2. **G2's conclusion is partly in Motegi, and so is its type-C twin's.** §8. A real finding, and against me.

3. **Steps 1–3 do not "transfer"; they are the same steps**, for a reason worth isolating (Lemma 5). §3.

4. **The pre-registration on ρ^α holds** and the named risk — long and short exchanging between the types — is explained away rather than merely avoided. §4, §7.

5. **A pre-registered claim of mine about $\varepsilon = -1$ was refuted**, and the replacement is sharper than the claim. §6.

6. **One unexpected structural fact**: ψ_B and ψ_C come from different blocks. §5.

2 Setup

Notation follows `2026-10-18-c2-gauge-theorem-and-q-divisors.tex`, including the corrected divisor nomenclature: divisors are named by the *root* that cuts them, never by a root length.

Let $R := \mathbb{C}(q)[w_1^{\pm}, \dots, w_N^{\pm}]$, $F := \operatorname{Frac} R$, $P := 2^{[N]}$ identified with the T-fixed points of $LG(N, 2N)$. Motegi's symmetrisation formula (`2610.15344`, eq. `Motherfunction`, L473–480) is

$$F_{N,n,r}(u \mid w \mid \mathbf{I}) = \frac{1}{(n-r)!}\Big(\prod_{k=1}^{n} u_k^{-(N+n)}\Big) \sum_{\sigma \in S_n} T\big(u_{\sigma(1)}, \dots, u_{\sigma(n)}\big), \tag{1}$$

where, writing $v_j := u_{\sigma(j)}$ and calling slots $1, \dots, r$ **heads** and $r+1, \dots, n$ **tails**,

$$\begin{aligned} T(v) = {} & \underbrace{\prod_{j=1}^{r}\Big[(1-q)(v_j^2-1)\prod_{k=i_j+1}^{N}(v_j - qw_k)\prod_{k=1}^{i_j-1}(v_j - w_k)\Big]}_{\text{line 1}} \cdot \underbrace{\prod_{1\le j<k\le r}\frac{qv_j - v_k}{v_j - v_k}(v_jv_k - 1)}_{\text{line 2a}} \\ & \cdot \underbrace{\prod_{j>r}\prod_{k=1}^{N}(v_j - qw_k)}_{\text{line 2b}} \cdot \underbrace{\prod_{j>r}\prod_{k\le r}\frac{qv_j - v_k}{v_j - v_k}(v_jv_k - 1)}_{\text{line 3a}} \cdot \underbrace{\prod_{r<j<k\le n}(v_jv_k - q)}_{\text{line 3b}} \cdot \underbrace{\prod_{j>r}(\beta v_j - \gamma)}_{\text{line 3c}}. \end{aligned} \tag{2}$$

The two specialisations (L494–501) are

$$W^C_{N,n,r} := F\big|_{\beta=1,\gamma=0}, \qquad W^B_{N,n,r} := \Big(\prod_{k=1}^{n}\big(1 + u_k^{-1}\big)^{-1}\Big)F\big|_{\beta=1,\gamma=-1}, \tag{3}$$

and the normalisers (L1057–1062, L1361–1367) are

$$\mathcal{E}^C_{N,n,r}(u,w) = \frac{\overbrace{\prod_{j,k=1}^{n}(1 - qu_j^{-1}u_k)}^{Q_1} \cdot \prod_{j=1}^{n}\prod_{k=1}^{N}(1 - qu_jw_k)}{\underbrace{\prod_{1\le j<k\le n}(1-qu_ju_k)}_{Q_2} \cdot \prod_{1\le j\le k\le N}(1 - qw_jw_k) \cdot (1-q)^{n-r}}, \tag{4}$$

$$\mathcal{E}^B_{N,n,r}(u,w) = \frac{Q_1 \cdot \prod_{j=1}^{n}\prod_{k=1}^{N}(1 - qu_jw_k) \cdot \boxed{\prod_{j=1}^{n}(1-qu_j)}}{Q_2 \cdot \boxed{\prod_{j=1}^{n}(1-qu_j^2)} \cdot \prod_{1\le j<k\le N}(1 - qw_jw_k) \cdot \prod_{k=1}^{N}(1-qw_k) \cdot (1-q)^{n-r}}. \tag{5}$$

Write $\widetilde{A}^X_{N,n,r} := W^X_{N,n,r}/\mathcal{E}^X_{N,n,r}$ (Motegi's $\widetilde{WX}$) and, for a multiset $\mathbf{U}$ of values, $\widetilde{A}^X(\mathbf{U}) := \widetilde{A}^X_{N,|\mathbf{U}|,r}(\mathbf{U} \mid w \mid \mathbf{I})$. All of (1)–(5) was re-transcribed first-hand this session from the e-print source at the line numbers given; the five local copies of that source agree (md5 `c15a23f2`, 5/5).

Remark 1 (the two boxed blocks)**.** $\mathcal{E}^B$ and $\mathcal{E}^C$ differ in four places, but two of them — the w-side $\prod_{j<k}$ versus $\prod_{j\le k}$, and the extra $\prod_k(1-qw_k)$ — are independent of n and of u, hence cancel identically in any ratio $\mathcal{E}^X_n/\mathcal{E}^X_s$ at fixed w. The only surviving difference is the pair of boxed u-diagonal blocks. Measured, symbolically and for all $\mathbf{M}$ at $N = 2, 3$: $\mathcal{E}^B/\mathcal{E}^C = \prod_{k\notin\mathbf{M}}(1-qw_k^2)/(1-qw_k)$ under $u = w_{\mathbf{M}}$, 28/28.

For $\mathbf{M} \in P$ put $\widetilde{A}^X_{\mathbf{M},\mathbf{I}} := \widetilde{A}^X(w_{\mathbf{M}})$ with $w_{\mathbf{M}} = \{w_m : m \in \mathbf{M}\}$ and $n = s := |\mathbf{M}|$; the absence of trailing 1's is legitimate in type B by *Motegi's own type-B* iterated stability, L3602–3608 (for $s \ge r$), and $\widetilde{A}^B_{\mathbf{M},\mathbf{I}} = 0$ for $s < r$ is L3620–3626. I flag this because the corresponding type-C citations, L2822–2828 and L2843, are the ones carried in my engine's docstring, and reusing a type-C locator for a type-B claim is exactly the 2026-10-17 defect.

Definition 2 (the collision data of a root of B_N)**.** The positive roots of B_N are $e_i - e_j$, $e_i + e_j$ $(i < j)$ and e_m. For each, H_α is the vanishing locus of the corresponding T-curve character, s_α the signed-permutation action on P, and $D(\alpha)$ the set of labels deleted:

root	H_α	$s_\alpha(\mathbf{M})$ on the colliding stratum	$D(\alpha)$
$e_i - e_j$	$\{w_i = w_j\}$	$\mathbf{M}\triangle\{i,j\}$ $(\lvert\mathbf{M}\cap\{i,j\}\rvert = 1)$	$\varnothing$
e_m	$\{w_m = 1\}$	$\mathbf{M} \setminus \{m\}$ $(m \in \mathbf{M})$	$\{m\}$
$e_a + e_b$	$\{w_a w_b = 1\}$	$\mathbf{M} \setminus \{a,b\}$ $(\{a,b\} \subseteq \mathbf{M})$	$\{a,b\}$

This is Definition `def:D` of `2026-10-18-c2` at $h = 1$: the boundary root of C_N is $2e_m$ with character w_m^2, of B_N is e_m with character w_m. Lemma `lem:model` there — that this is the signed-permutation action — is a statement about $W(C_N) = W(B_N)$ and is type-independent.

Set $\iota(v) = v^{-1}$. A finite multiset $\mathbf{D} \subset F^\times$ is **ι-closed** if some involution τ of its index set satisfies $y_{\tau(j)} = y_j^{-1}$; the τ-fixed points are the y with $y^2 = 1$.

Lemma 3 (the three root families of B_N are the three ι-orbit types)**.** *Let $\alpha > 0$ in B_N, $\mathbf{M}$ in its colliding stratum, and $\mathbf{D}(\alpha) := \{w_m : m \in D(\alpha)\}$ restricted to H_α. Then $\mathbf{D}(\alpha)$ is ι-closed, and*

$$\alpha = e_i - e_j \;\Rightarrow\; \mathbf{D} = \varnothing; \quad \alpha = e_m \;\Rightarrow\; \mathbf{D} = \{1\}\ (\tau\text{-fixed}); \quad \alpha = e_a + e_b \;\Rightarrow\; \mathbf{D} = \{x, x^{-1}\}\ (\text{free } \tau\text{-orbit}),$$

and on H_α the value multisets satisfy $w_{\mathbf{M}} = w_{s_\alpha\mathbf{M}} \sqcup \mathbf{D}(\alpha)$.

Proof. Verbatim the proof of Lemma `lem:orbits` of `2026-10-18-c2`, with one simplification: where type C's boundary locus $\{w_m^2 = 1\}$ gives $\varepsilon = w_m$ with $\varepsilon = \pm 1$, type B's locus $\{w_m = 1\}$ gives $\varepsilon = 1$ outright. The A-type clause is that on $\{w_i = w_j\}$ with $i \in \mathbf{M} \not\ni j$ the two value multisets are *equal*. □

Remark 4 (the one structural asymmetry between the types)**.** $\{w_m^2 = 1\}$ has the two components $w_m = \pm 1$; $\{w_m = 1\}$ has one. Both $+1$ and -1 are τ-fixed points of ι in either type, but only in type C are both of them *root* hyperplanes. §6 shows that this is not a cosmetic difference: it is exactly where the type-B theorem's domain stops.

3 Steps 1–3 are not analogous to type C's; they are identical

Lemma 5 (the type comparison, with the prefactor cancelled analytically)**.** *Fix $\sigma \in S_n$ and write T_σ^X for the summand of (1) contributing to W^X, so that $W^X = \frac{1}{(n-r)!}\big(\prod_k u_k^{-(N+n)}\big)\sum_\sigma T_\sigma^X$ with the convention that W^B's prefactor is distributed over the summands. Then, as rational functions,*

$$T_\sigma^B = T_\sigma^C \cdot \prod_{j=1}^{r} \frac{v_j}{v_j+1},$$

and explicitly T_σ^B equals (2) with line 3c deleted, line 1's head factor $(1-q)(v_j^2-1)$ replaced by

$$\boxed{(1-q)(v_j-1)}$$

and an overall $\prod_{k=1}^n u_k$. In particular T_σ^B has no pole at $u_k=-1$.

Proof. $\prod_{k=1}^n (1+u_k^{-1})^{-1} = \prod_{k=1}^n u_k/(u_k+1)$, and line 3c at $\beta=1,\gamma=-1$ is $\prod_{j>r}(v_j+1)$. Since $\prod_{k=1}^n(u_k+1) = \prod_{j=1}^n(v_j+1)$ — the same multiset of values — we get

$$\prod_{k=1}^{n} \frac{u_k}{u_k+1} \cdot \prod_{j>r}(v_j+1) = \frac{\prod_{k=1}^n u_k}{\prod_{j\le r}(v_j+1)},$$

so $T_\sigma^B = [\text{lines } 1, 2a, 2b, 3a, 3b] \cdot \prod_k u_k \big/ \prod_{j\le r}(v_j+1)$. Absorbing $1/(v_j+1)$ into line 1 via $v_j^2-1 = (v_j-1)(v_j+1)$ gives the boxed head factor and the displayed explicit form. Comparing with $T_\sigma^C = [\text{lines } 1, 2a, 2b, 3a, 3b] \cdot \prod_{j>r} v_j$ and using $\prod_k u_k = \prod_{j\le r} v_j \cdot \prod_{j>r} v_j$ gives the ratio. No factor of the explicit form has u_k+1 in a denominator, whence the last claim. □

Remark 6 (why this cancellation has to be done before substituting)**.** W^B's prefactor has a *pole* at $u_k=-1$ and line 3c a *zero* there, so the literal expression (3) evaluates to $\infty\cdot 0$ at $u_k=-1$: undefined, not wrong. I ran the type-independence census on the literal form first and it returned 0/18 at $\varepsilon=-1$, which I nearly read as a refutation of Lemma 11. It was an undefined computation. The correct reading of 0/18 was "you substituted into a removable singularity"; and when I then did the cancellation, the real phenomenon turned out to be §6, which is a different and sharper statement. Both the false reading and the true one live at the same point of the same locus.

Lemma 7 (head exclusion in type B, and exactly where it fails)**.** *Let $\mathbf{D}$ be ι-closed with realising involution τ, and let the values of $\mathbf{R}\sqcup\mathbf{D}$ be **pairwise distinct**. Let σ place slot p of $\mathbf{D}$ in a head position. Then $T_\sigma^B = 0$ if either*

(a) $\tau(p)\neq p$, or

(b) $\tau(p)=p$ and $u_p=+1$;

and if $\tau(p)=p$ with $u_p=-1$ then $T_\sigma^B\neq 0$ in general. In type C, both (a) and $\tau(p)=p$ with $u_p=\pm 1$ give $T_\sigma^C=0$.

Proof. (a) The slot pair $\{p,\tau(p)\}$ contains the head p, so it carries the factor $(u_p u_{\tau(p)}-1)$ from line 2a (if $\tau(p)$ is also a head) or line 3a (if it is a tail); and $u_p u_{\tau(p)}=1$. These two lines are unchanged by Lemma 5, so this case is type-independent. Distinctness enters exactly here: the

same slot pair carries the *denominator* $(v_p - v_{p'})$ in lines 2a/3a, and if two values of $\mathbf{D}$ coincide the term is $0/0$ rather than 0 (Remark `rem:distinct` of `2026-10-18-c2`).

(b) and the failure. By Lemma 5 the head factor at slot p is $(1-q)(u_p-1)$ in type B and $(1-q)(u_p^2-1)$ in type C. A τ-fixed point has $u_p^2 = 1$, i.e. $u_p = \pm 1$. Then $u_p^2 - 1 = 0$ always, while $u_p - 1 = 0$ iff $u_p = +1$ and equals -2 at $u_p = -1$; and no other factor of the explicit form of Lemma 5 vanishes at $u_p = -1$ for generic $\mathbf{R}$ and w. □

Remark 8 (measured)**.** Lemma 7's split is exactly what the head-slot term does: at $\varepsilon = +1$ it vanishes in 9/9 instances ($N = 2, 3$, $r = 1, 2$, every $\mathbf{I}$), and at $\varepsilon = -1$ it fails to vanish in 9/9. A clean split on the nose of the stated discriminator.

Definition 9 (admissible)**.** Call an ι-closed $\mathbf{D}$ **X-admissible** if its values are pairwise distinct and, when $X = B$, no τ-fixed point of $\mathbf{D}$ equals -1. (For $X = C$ no extra condition.) When $r = 0$ every ι-closed $\mathbf{D}$ with distinct values is admissible in both types, since there is then no head slot.

Corollary 10 (the head selections match)**.** *Let $\mathbf{D}$ be X-admissible, $s = |\mathbf{R}|$, $d = |\mathbf{D}|$, $n = s + d$. Then*

$$\frac{1}{(n-r)!}\sum_{\sigma\in S_n} T_\sigma^X = \sum_{\substack{(p_1,\dots,p_r)\ \text{distinct}\\ \text{slots of } \mathbf{R}}} T^X\big(p_1,\dots,p_r;\ \text{tails} = \text{the rest}\big),$$

and the index set on the right is the same for $\mathbf{R}\sqcup\mathbf{D}$ and for $\mathbf{R}$.

Proof. Tail-symmetry: lines 2b, 3b are tail-symmetric, line 3a is a product over tails of head-dependent factors, and the extra data of Lemma 5 — the overall $\prod_k u_k$ and the head factors — is symmetric in the tails. So the $(n-r)!$ tail orderings give equal terms. By Lemma 7, with $\mathbf{D}$ admissible, the surviving head selections avoid $\mathbf{D}$. □

Lemma 11 (the unnormalised adjunction ratio is type-independent)**.** *Let $\mathbf{D}$ be B- and C-admissible and $s = |\mathbf{R}| \geq r$. Then*

$$\frac{W^B_{N,n,r}(\mathbf{R}\sqcup\mathbf{D})}{W^B_{N,s,r}(\mathbf{R})} = \frac{W^C_{N,n,r}(\mathbf{R}\sqcup\mathbf{D})}{W^C_{N,s,r}(\mathbf{R})}.$$

Proof. By Corollary 10 both numerator and denominator are sums over the *same* head selections $\mathbf{h}\subset\mathbf{R}$. Fix one. By Lemma 5 the type-B summand is the type-C summand times $\prod_{v\in\mathbf{h}} v/(v+1)$, a factor depending only on $\mathbf{h}$ — and in particular not on $\mathbf{D}$. It therefore cancels between numerator and denominator term by term. □

Remark 12. This is stronger than yesterday's gap statement, which read "Steps 1 and 2 are untouched; what changes is the bookkeeping of Steps 3–5". Steps 1, 2 *and* 3 are untouched: the whole F-side computation is the type-C one, and only the normaliser has to be redone. I had located the type-dependence one step too early — and, per Lemma 7, one *hypothesis* too weakly.

So I may quote eq. `eq:Fratio` of `2026-10-18-c2` as it stands. Recall

$$Q_1(\mathbf{D}) = \prod_{\text{ordered } (j,k)}\Big(1 - q\tfrac{y_k}{y_j}\Big),\quad Q_2(\mathbf{D}) = \prod_{\{j,k\},\,j\neq k}(1-qy_jy_k),\quad P_2(\mathbf{D}) = \prod_{\{j,k\},\,j\neq k}(y_jy_k - q),$$

and the two reindexing identities, proved there for arbitrary ι-closed $\mathbf{D}$ and *type-free* (they are statements about a multiset with an involution; no weight function occurs in either):

$$\text{(I1)}\quad P_2(\mathbf{D}) = \Big(\prod_{y\in\mathbf{D}} y\Big)^{d-1} Q_2(\mathbf{D}), \qquad \text{(I2)}\quad \frac{Q_2(\mathbf{D})^2}{Q_1(\mathbf{D})} = \frac{1}{\prod_{y\in\mathbf{D}}(1-qy^2)}. \tag{6}$$

Proposition 13 (Step 3, quoted)**.** *With* $\mathbf{D}$ *admissible,* $s \geq r$*,* $n = s + d$*, and either type,*

$$\frac{W^X_{N,n,r}(\mathbf{R}\sqcup\mathbf{D})}{W^X_{N,s,r}(\mathbf{R})} = \Big(\prod_{y\in\mathbf{D}} y\Big)^{1-d} P_2(\mathbf{D}) \prod_{y\in\mathbf{D}} \Big[\prod_{v\in\mathbf{R}} \big(1-\tfrac{qy}{v}\big) \prod_{k=1}^{N} (1-qyw_k)\Big]. \tag{7}$$

Proof. For $X = C$ this is Steps 1–3 of Theorem `thm:gauge` in `2026-10-18-c2`; for $X = B$ it follows by Lemma 11. □

4 Step 4: the normaliser, and the theorem

Lemma 14 (Step 4, type B)**.** *With* $\mathbf{D}$ *admissible and* $s \geq r$*,*

$$\frac{\mathcal{E}^B_{N,n,r}}{\mathcal{E}^B_{N,s,r}} = \frac{Q_1(\mathbf{D})}{Q_2(\mathbf{D})\,(1-q)^d} \cdot \frac{\prod_{y\in\mathbf{D}}(1-qy)}{\prod_{y\in\mathbf{D}}(1-qy^2)} \cdot \prod_{y\in\mathbf{D}} \Big[\prod_{v\in\mathbf{R}} \big(1-\tfrac{qy}{v}\big) \prod_{k=1}^{N} (1-qyw_k)\Big]. \tag{8}$$

Proof. Both normalisers sit at the same alphabet w, so by Remark 1 the two w-only blocks cancel. Adjoining $\mathbf{D}$ changes each remaining block thus.

Q_1: acquires the $\mathbf{DD}$ pairs $Q_1(\mathbf{D})$ and the mixed pairs $\prod_{y\in\mathbf{D}}\prod_{v\in\mathbf{R}}\big(1-q\frac{v}{y}\big)\big(1-q\frac{y}{v}\big)$, numerator.
$\prod_j\prod_k(1-qu_jw_k)$: acquires $\prod_y\prod_k(1-qyw_k)$, numerator.
Q_2: acquires $Q_2(\mathbf{D})\prod_y\prod_{v\in\mathbf{R}}(1-qyv)$, denominator.
$(1-q)^{n-r}$: acquires $(1-q)^d$, denominator.
The boxed blocks: $\prod_j(1-qu_j)$ acquires $\prod_y(1-qy)$ in the numerator, and $\prod_j(1-qu_j^2)$ acquires $\prod_y(1-qy^2)$ in the denominator.

Finally, since τ is an involution with $y_{\tau(j)} = y_j^{-1}$, $\prod_y\prod_v\big(1-q\frac{v}{y}\big) = \prod_y\prod_v(1-qvy)$, which cancels the denominator's $\prod_y\prod_v(1-qyv)$. Collecting gives (8). □

Theorem 15 (the type-B ι-adjunction law)**.** *Let* $\mathbf{R}$ *be a finite multiset of values with* $s = |\mathbf{R}| \geq r = |\mathbf{I}|$ *and let* $\mathbf{D}$ *be* ι*-closed,* $d = |\mathbf{D}|$*, and* B***-admissible*** *(Definition 9): values of* $\mathbf{R}\sqcup\mathbf{D}$ *pairwise distinct, and no* τ*-fixed point of* $\mathbf{D}$ *equal to* -1 *unless* $r = 0$*. Then, at a generic point of the locus on which* $\mathbf{D}$ *is* ι*-closed,*

$$\boxed{\widetilde{A}^B(\mathbf{R}\sqcup\mathbf{D}) = \Big(\prod_{y\in\mathbf{D}} \varphi_B(y)\Big)\widetilde{A}^B(\mathbf{R}), \qquad \varphi_B(y) = \frac{1-q}{1-qy}}$$

Proof. Divide (7) by (8); the bracketed products are identical and cancel, leaving

$$\frac{\widetilde{A}^B(\mathbf{R}\sqcup\mathbf{D})}{\widetilde{A}^B(\mathbf{R})} = \Big(\prod_y y\Big)^{1-d} P_2(\mathbf{D}) \cdot \frac{Q_2(\mathbf{D})\,(1-q)^d}{Q_1(\mathbf{D})} \cdot \frac{\prod_y(1-qy^2)}{\prod_y(1-qy)}.$$

By (I1), $\big(\prod_y y\big)^{1-d} P_2(\mathbf{D}) = Q_2(\mathbf{D})$, so the first three factors are $(1-q)^d Q_2(\mathbf{D})^2/Q_1(\mathbf{D})$, which by (I2) is $(1-q)^d/\prod_y(1-qy^2)$. Hence

$$\frac{\widetilde{A}^B(\mathbf{R}\sqcup\mathbf{D})}{\widetilde{A}^B(\mathbf{R})} = \frac{(1-q)^d}{\prod_y(1-qy^2)} \cdot \frac{\prod_y(1-qy^2)}{\prod_y(1-qy)} = \prod_{y\in\mathbf{D}} \frac{1-q}{1-qy}.$$

□

Corollary 16 (G2: ρ^α for every root of B_N, all N)**.** *Set* $\psi_B(v) = (1-qv)/(1-q)$, $\Psi_B(\mathbf{M}) = \prod_{m\in\mathbf{M}} \psi_B(w_m)$ *and* $\widehat{A}^B_{\mathbf{M},\mathbf{I}} = \widetilde{A}^B_{\mathbf{M},\mathbf{I}}\Psi_B(\mathbf{M})$. *For every* N, *every positive root* α *of* B_N, *every* $\mathbf{M}$ *in the colliding stratum of* α *and every* $\mathbf{I}$, *on* H_α

$$\widetilde{A}^B_{\mathbf{M},\mathbf{I}} = \rho^\alpha\, \widetilde{A}^B_{s_\alpha\mathbf{M},\mathbf{I}}, \qquad \rho^\alpha = \prod_{m\in D(\alpha)} \varphi_B(w_m) = \prod_{m\in D(\alpha)} \psi_B(w_m)^{-1},$$

and consequently $\widehat{A}^B_{\mathbf{M},\mathbf{I}} = \widehat{A}^B_{s_\alpha\mathbf{M},\mathbf{I}}$ *on* H_α*: the collapse multiplier of* $\widehat{A}^B$ *is exactly* 1 *at every root of* B_N, *for every* N.

Proof. Put $\mathbf{R} = w_{s_\alpha\mathbf{M}}$, $\mathbf{D} = \mathbf{D}(\alpha)$, so $w_\mathbf{M} = \mathbf{R}\sqcup\mathbf{D}$ with $\mathbf{D}$ ι-closed (Lemma 3). Suppose $|\mathbf{R}| \geq r$. $\mathbf{D}(\alpha)$ *is* B*-admissible for every root of* B_N: its values are $\varnothing$, $\{1\}$ or $\{x, x^{-1}\}$, so its only possible τ-fixed point is $+1$ — never -1 — and distinctness holds at the generic point of H_α (the values of $\mathbf{R}$ are distinct indeterminates; $|\mathbf{D}| \leq 2$; and when $|\mathbf{D}| = 2$ its values satisfy $w_a w_b = 1$ and so coincide only on $\{w_a^2 = 1\} \cap H_{e_a+e_b}$, a proper closed subvariety). This is the point at which Lemma 7's restriction is seen not to bite: *no root of* B_N *produces* -1 *as a deleted value*, precisely because $\{w_m = -1\}$ is not a root hyperplane of B_N (Remark 4). Theorem 15 then gives the multiplier at the generic point; both sides lie in $\mathcal{O}_{H_\alpha}$, so the identity holds in the local ring. Evaluating φ_B:

α	$\lvert D\rvert$	$\mathbf{D}$ on H_α	ρ^α
$e_i - e_j$	0	$\varnothing$	1
e_m	1	$\{1\}$	$\varphi_B(1) = \dfrac{1-q}{1-q} = 1$
$e_a + e_b$	2	$\{x, x^{-1}\}$	$\dfrac{(1-q)^2}{(1-qx)(1-qx^{-1})}$

If $|\mathbf{R}| < r$ both sides vanish: Proposition 18. For $\widehat{A}^B$: $\mathbf{M}$ and $s_\alpha\mathbf{M}$ differ exactly in $D(\alpha)$, so $\Psi_B(\mathbf{M}) = \Psi_B(s_\alpha\mathbf{M})\prod_{m\in D(\alpha)}\psi_B(w_m)$ and the two factors are mutually inverse. For $\alpha = e_i - e_j$, $D = \varnothing$ and $\Psi_B(\mathbf{M}) = \Psi_B(s_{ij}\mathbf{M})$ on $\{w_i = w_j\}$ since $\psi_B(w_i) = \psi_B(w_j)$ there. □

Remark 17 (the $d = 0$ case is u-symmetry, not an instance of the box)**.** At $\mathbf{D} = \varnothing$ Theorem 15 reads $\widetilde{A}^B(\mathbf{R}) = \widetilde{A}^B(\mathbf{R})$, vacuous as an identity; the A-type content sits entirely in the second clause of Lemma 3, that the two value multisets coincide on H_α. As in type C I say this rather than claim that all three families are instances of one identity.

Proposition 18 (the vacuous strata vanish by head exclusion, not by triangularity)**.** *Let* $\alpha > 0$, $\mathbf{M}$ *in its colliding stratum,* $\mathbf{R} = w_{s_\alpha\mathbf{M}}$ *and* $r = |\mathbf{I}| > |\mathbf{R}|$. *Then on* H_α *both* $\widetilde{A}^X_{s_\alpha\mathbf{M},\mathbf{I}} = 0$ *and* $\widetilde{A}^X_{\mathbf{M},\mathbf{I}} = 0$, *and Corollary 16 reads* $0 = \rho^\alpha \cdot 0$.

Proof. $\widetilde{A}^X_{s_\alpha\mathbf{M},\mathbf{I}} = 0$ is lower-cardinality vanishing ($|s_\alpha\mathbf{M}| < r$): L3620–3626 in type B, L2843 in type C. For $\widetilde{A}^X_{\mathbf{M},\mathbf{I}}$: by Lemma 7 and Corollary 10 the surviving head selections are the r-element subsets of the slots of $\mathbf{R}$, of which there are none when $r > |\mathbf{R}|$; the sum (1) is empty. □

Retraction 19 (a wrong citation in my own type-C paper)**.** Corollary `cor:heads` of `2026-10-18-c2` disposes of this case with *"(If $s < r$ both sides vanish: $\widetilde{A}_{\mathbf{M},\mathbf{I}} = \widetilde{A}_{\mathbf{R},\mathbf{I}} = 0$ by (M3)…)"*. For $\widetilde{A}_{\mathbf{R},\mathbf{I}}$ that is right. For $\widetilde{A}_{\mathbf{M},\mathbf{I}}$ it is **wrong**: $|\mathbf{M}| = |\mathbf{R}| + d$, so in the pair family with $|\mathbf{R}| = r - 1$ one has $|\mathbf{M}| = r + 1 > r$ and (M3)'s cardinality clause does not apply, while its triangularity clause $m_{|\mathbf{M}|-r+j} \geq i_j$ is generally satisfied — e.g. $N = 2$, $\mathbf{M} = \{1, 2\}$, $\mathbf{I} = \{1\}$, where $m_2 = 2 \geq 1$. The conclusion is correct and the reason given is not; the correct reason is head exclusion. I single this

out because it is the characteristic shape of my errors this month: a true statement resting on a cited authority that does not cover it, and no objector, because the conclusion is right. Measured here: 124/124 instances at $N = 2, 3$ over all nine roots vanish as Proposition 18 says.

5 Where the two ψ's come from, and why that is not one mechanism

Yesterday I answered Rick's question *"where does ψ come from on Motegi's side?"* with (I2): $\varphi = \psi^{-1}$ is $(1-q)/(1-qy^2)$ because $Q_2^2/Q_1 = \prod_y (1-qy^2)^{-1}$, the identity holding because τ matches $\mathcal{E}^C$'s $\prod_{j,k}(1-qu_j^{-1}u_k)$ against its $\prod_{j<k}(1-qu_ju_k)$; and I described this as "ψ is a ratio of two blocks of the normaliser, read along ι". Theorem 15 shows that answer is **type-C-specific**. Compare the Step-5 computations block by block:

	type C	type B
(I1)+(I2) output	$(1-q)^d/\prod_y(1-qy^2)$	$(1-q)^d/\prod_y(1-qy^2)$
$\mathcal{E}^X$'s u-diagonal blocks	*none*	$\prod_y(1-qy^2)/\prod_y(1-qy)$
$\prod_y \varphi_X(y)$	$(1-q)^d/\prod_y(1-qy^2)$	$(1-q)^d/\prod_y(1-qy)$

In type B the output of (I2) — the very factor that *is* ψ_C — is cancelled **exactly** by $\mathcal{E}^B$'s $\prod_j(1-qu_j^2)$, and ψ_B is instead $\mathcal{E}^B$'s numerator block $\prod_j(1-qu_j)$, read along ι.

Corollary 20 (the type-dependence of ρ, isolated)**.** *For a positive root α of B_N resp. C_N with the same collision data $D(\alpha)$,*

$$\frac{\rho_B^\alpha}{\rho_C^\alpha} = \prod_{m\in D(\alpha)} \frac{1-qw_m^2}{1-qw_m},$$

which is precisely the ratio of the boxed u-diagonal blocks of (5) and (4). More generally, for **D** *admissible in both types, $\prod_y \varphi_B(y)/\prod_y \varphi_C(y) = \prod_{y\in\mathbf{D}}(1-qy^2)/(1-qy)$.*

Proof. By Lemma 11 the W-ratios agree, so the quotient of the multipliers is the inverse quotient of the $\mathcal{E}^X$-ratios, which by Remark 1 and Lemma 14 is the boxed-block ratio. □

Remark 21 (the honest revision of my answer to G5)**.** The uniform formula $\psi_X(v) = (1-qv^h)/(1-q)$ is correct and it is a *coincidence of outputs*, not one derivation with a parameter. No step of either proof has h as a variable. What there is: (a) a type-free identity (I2) with output $\prod(1-qy^2)^{-1}$, and (b) the u-diagonal blocks the normaliser happens to carry. In type C (a) survives and (b) is empty; in type B (b) annihilates (a) and supplies the answer itself. So the general statement of "where ψ comes from" is *the u-diagonal content of $\mathcal{E}^X$ after the ι-pairing*, which reduces to (I2) only when $\mathcal{E}^X$ has no u-diagonal block of its own. And h turns out to live in *two* places, not one: also in the head factor (v^h-1) of Lemma 5, which is what §6 is about.

6 A pre-registered claim of mine, refuted; and the domain is the root system

Proposition 27 below shows Motegi's propositions already cover **D** of the two shapes $\{1\}$ and $\{x, x^{-1}\}$. I pre-registered, when drafting this paper, that Theorem 15's extra content would be visible at the second τ-fixed point:

*(pre-registered, and **false**)* $\mathbf{D} = \{-1\}$ is ι-closed, so $\widetilde{A}^B(\mathbf{R} \sqcup \{-1\}) = \frac{1-q}{1+q}\widetilde{A}^B(\mathbf{R})$ whereas $\widetilde{A}^C(\mathbf{R} \sqcup \{-1\}) = \widetilde{A}^C(\mathbf{R})$; this is the sharp type-discriminator, in neither of Motegi's propositions.

Theorem 22 (the B-admissibility hypothesis is necessary)**.** *For $\mathbf{D} = \{-1\}$ and $r \geq 1$ the conclusion of Theorem 15 is **false**. For $r = 0$ it holds.*

Proof. For $r = 0$ there is no head slot, Lemma 7 is vacuous, and the proof of Theorem 15 goes through: $\mathbf{D}$ is admissible by Definition 9. For $r \geq 1$, Lemma 7 shows that the σ placing -1 in a head slot contributes a nonzero term, so Corollary 10's index sets do *not* match and Lemma 11 is unavailable. That is an obstruction to the proof, not to the statement; the statement is refuted by computation. Measured with the cancelled form of Lemma 5, symbolic in w and q, over $N = 2, 3$ and all $\mathbf{I}$: $r = 0$**:** $6/6$ **agree with** $\varphi_B(-1)$**;** $r \geq 1$**:** $0/9$, and the measured ratio is not even a function of q alone — at $N = 2$, $s = 1$, $\mathbf{I} = \{1\}$ it is

$$\frac{18q^3w_1 - 45q^2w_1 - 108q^2 + 27qw_1 + 45q + 63}{10q^3w_1 + 45q^2w_1 + 10q^2 + 35qw_1 + 45q + 35},$$

whereas $\varphi_B(-1) = (1-q)/(1+q)$ depends on no w at all. So the failure is not a different scalar; the adjunction is not by a scalar. □

Remark 23 (the shape of this refutation is one I have seen before)**.** A hypothesis of Lemma 7 was discovered by its own failure mode, and **the discriminator is a head slot — the discriminator the lemma itself names**. That is verbatim the structure of Remark `rem:distinct` of `2026-10-18-c2`, where the omitted distinctness hypothesis was refuted by a census that was green at $r = 0$ and failed at $r \geq 1$. Two different hypotheses of the same lemma, found the same way, one day apart. The lesson I draw is not about either hypothesis: it is that $r = 0$ *is a degenerate stratum which silently satisfies every head hypothesis*, so a census that is green at $r = 0$ and never run at $r \geq 1$ carries no information about head exclusion at all.

What replaces the pre-registered claim is better, because it says why -1 had to be the place where this happens.

Corollary 24 (the theorem's domain is cut out by its own root system)**.** *In type X the τ-fixed points at which head exclusion fires are exactly the h_X-th roots of unity: $\{\pm 1\}$ for $h = 2$ (type C), $\{+1\}$ for $h = 1$ (type B). These are exactly the points of the boundary root hyperplane $\{w_m^h = 1\}$ of that type. Hence*

1. *every root hyperplane of B_N delivers an admissible $\mathbf{D}$ (Corollary 16), so G2 is untouched by Theorem 22;*
2. *the type-B theorem reaches no τ-fixed point beyond its own root system, and the type-C theorem reaches both components of $\{w_m^2 = 1\}$ — which is content beyond `stabilityWC`, since that proposition is stated at $u_n = 1$ only.*

Proof. A τ-fixed point satisfies $y^2 = 1$. By Lemma 5 the head factor is $(1-q)(v^h - 1)$ with $h = h_X$ — $(v^2 - 1)$ in type C, $(v - 1)$ in type B — whose zero set among $\{y : y^2 = 1\}$ is the set of h-th roots of unity; and $\{w_m^h = 1\}$ is by Definition 2 the boundary root hyperplane. (1) is the admissibility computation in Corollary 16. (2) is Lemma 7's type-C clause. □

Remark 25 (so what *is* new beyond Motegi's propositions)**.** Not the second fixed point in type B: that is outside the domain. What remains, and is genuine:

- $d \geq 3$ **and multi-orbit D**, within the admissible domain: e.g. $\mathbf{D} = \{x, x^{-1}, y, y^{-1}\}$, where Motegi's $d = 2$ proposition applies only after being iterated, and $\mathbf{D} = \{1, x, x^{-1}\}$, mixing orbit types. Measured green.
- **The second component** $\{w_m = -1\}$ **in type** C, which is a root hyperplane there and is *not* $u_n = 1$.
- **Both the unification and the obstruction**: that the three families are the three ι-orbit types (hence three and no more), and that the domain is cut out by h in the head factor (Corollary 24). Neither is visible from two separate propositions.

7 The named risk: long and short exchange, and nothing happens

The brief flagged a specific danger. The root $e_a + e_b$ is **long in** B_N **and short in** C_N (verified from $|\alpha|^2$ at $n = 2, 3, 4$ on 10-18; the naming defect I corrected in 35 places), while the boundary root is $2e_m$ (long) in C_N and e_m (short) in B_N. If the orbit type attaches to the *root* and the multiplier to the *length*, the two types must disagree about which family gets which multiplier, and the type-C theorem's vocabulary breaks.

It does not happen, and the proof says why. Reading Theorem 15, Lemma 5 and Lemma 14: ρ^α is a function of exactly two inputs, (i) the ι-orbit decomposition of $\mathbf{D}(\alpha)$, and (ii) the exponent h, which appears in the head factor $(v^h - 1)$ and in the u-diagonal block of $\mathcal{E}^X$. *No step references a root length, or $|\alpha|^2$, or the long/short dichotomy.* The exchange of lengths is therefore invisible to the statement.

Remark 26 (so what is h?)**.** Worth saying once precisely, because my own `2026-10-17-typeB` Theorem `thm:rhoB` glosses it as "h the length of the boundary root: $2t_c$ in type C, t_c in type B", and that will mislead whoever reads it next. h_X is the exponent in the boundary *character* w_m^h — equivalently (Lemma 5, Corollary 20) the exponent in the head factor and in $\mathcal{E}^X$'s u-diagonal block. It is not a length: $|2e_m|^2 = 4$ and $|e_m|^2 = 1$, neither of which is h. It correlates with "long" within the boundary family by accident of there being two types; and the uniform formula applies h to the *pair* family too, where $e_a + e_b$ is long in B and short in C — the opposite ordering. The formula is right; the gloss is a quantifier error of the same kind as the long-root/short-root divisor naming, and I retract it here rather than leave it standing in a paper Rick has.

8 What is already in Motegi, including for type C

Task 0 of the brief was mandatory and it fired. Reading `2610.15344` §"Recursive relations and stability" for both types, first-hand:

Type B**, pair family.** `normalizedsusythree`, L2767–2771. *Assume* $r \leq n-2$. *Then* $\widetilde{WB}_{N,n,r}(\mathbf{u}|\mathbf{w}|\mathbf{I})\big|_{u_i u_j = 1} = \frac{(1-q)^2}{(1-qu_i)(1-qu_j)} \widetilde{WB}_{N,n-2,r}(\mathbf{u} \smallsetminus \{u_i, u_j\}|\mathbf{w}|\mathbf{I})$.

Type B**, boundary family.** `stabilityWB`, L2779–2783. *Assume* $r \leq n-1$. *Then* $\widetilde{WB}_{N,n,r}\big|_{u_n = 1} = \widetilde{WB}_{N,n-1,r}(\mathbf{u} \smallsetminus \{u_n\})$: multiplier 1.

Type C, pair family. `normalizedsusytwo`, L2618–2622, with a proof: multiplier $\frac{(1-q)^2}{(1-qu_i^2)(1-qu_j^2)}$.

Type C, boundary family. `stabilityWC`, L2643–2647: multiplier 1 at $u_n = 1$.

These are u-specialisations with $\mathbf{w}$ free and independent of $\mathbf{u}$; mine are w-specialisations with $\mathbf{u} = \mathbf{w_M}$ determined by $\mathbf{w}$. The implication runs one way, and easily:

Proposition 27 (Motegi's propositions imply the pair and boundary cases)**.** *Let $\alpha = e_a + e_b$ resp. e_m, $\mathbf{M} \supseteq D(\alpha)$, $r \le |s_\alpha \mathbf{M}|$. Then $(\mathbf{u}, \mathbf{w}) = (\mathbf{w_M}, \mathbf{w})$ with $\mathbf{w}$ generic in H_α lies on Motegi's locus $\{u_i u_j = 1\}$ resp. $\{u_n = 1\}$, with i, j resp. n the slots carrying $D(\alpha)$, and both sides of his identity are regular there; so his identity specialises to Corollary 16 for those two families, with $n = |\mathbf{M}|$.*

Proof. On $\{w_a w_b = 1\}$ the slots carrying w_a, w_b satisfy $u_i u_j = 1$; on $\{w_m = 1\}$ the slot carrying w_m has value 1. Regularity: the only denominators of (2) are the $(v_p - v_{p'})$ of lines 2a and 3a, nonvanishing because the values of $\mathbf{w_M}$ are pairwise distinct at the generic point of H_α; the W^B prefactor is regular and nonzero there by Lemma 5 (its only singularity is at $u_k = -1$, which is not on H_α for a root of B_N); and $\mathcal{E}^X$ has neither a zero nor a pole there, its factors being $1 - q \cdot$ (Laurent monomial) evaluated away from the corresponding (Q)-divisor. His hypotheses $r \le n-2$ resp. $r \le n-1$ are exactly $r \le |s_\alpha \mathbf{M}|$. □

Retraction 28 (priority: ρ^C is Motegi's, and my own paper does not say so)**.** `2026-10-18-c2` §`sec:priority` lists under "**Not new**" that $\rho^C = (1-q)^2/((1-qw_a^h)(1-qw_b^h))$ "is Rick's Lemma QC and my `2026-10-16-c2` Thm `collapse`". That attribution is **incomplete and I retract it**: by Proposition 27 in type C, the same number is Motegi's `normalizedsusytwo`, L2618–2622, proved there. The same applies to the type-C boundary multiplier (`stabilityWC`) and, as of today, to both non-trivial type-B multipliers, one of which is my own `2026-10-17` Theorem `thm:rhoB`.

The measurement of how I missed it is unflattering and worth recording. `stabilityWC` is L2645; my engine's `Atilde` docstring has cited its iterate, L2822–2828, since 2026-10-15. `normalizedsusytwo` is 27 **lines above the statement I was already citing**, in the same subsection, under a heading that names it. This is the third consecutive session in which something I proved, or filed as a gap, sat in a paper I hold at `deep-read` within a few lines of a locator I had already written down (10-18: G3 at L3302–3309, four lines from Rick's citation; 10-18: the Koncki table, reachable from a second bibitem key). The pattern is not "I have not read the paper". It is that **an extraction level records provenance depth, not coverage**, and the cheapest available instrument — reading the neighbourhood of a locator I am already using — is the one I keep not reaching for.

What today's proof still buys, stated without inflation. A self-contained derivation, from the symmetrisation formula, of the arbitrary-admissible-ι-closed law in type B, which specialises to both of his propositions; plus Remark 25's three items; plus the structural facts of §5 and Corollary 24, which no pair of separate propositions exhibits. Had I graded G2 a corollary at Task 0 and stopped — which the brief explicitly authorised — I would not have found Theorem 22 or Corollary 24, and I would still believe Remark 21's type-C answer was the general one.

9 Verification

All computation is pure SymPy (no combinatorics library), in `proofs/code-1019-typeB-gauge/`. The engine `q484_engine.py` is the 2026-10-18 copy, md5 `f53c025810f95130ccec70122b1e1ae3`,

identical across all five in-repository copies (verified as a set, not by reading one). The e-print source used for every transcription is md5 `c15a23f2...`, identical across all five local copies.

V3 — the engine, validated against type-B theorems not used to write it. The 2026-10-17 defect was an engine that normalised type B by $\mathcal{E}^C$ and produced a type-C arm that was green and structurally blind. The fix is to validate against a *different* theorem from the definition the code was written from. **Deliberately excluded** from this file: `stabilityWB` and `normalizedsusythree`, because those two *are* (specialisations of) today's theorem, and using them to validate the engine and then the engine to corroborate the theorem would be circular.

validator	Motegi locator	result
diagonal is a root product, type B	`typeBmainproperties`(i), L1386	12/12
w-symmetry $\widetilde{WB}(u\|w\|\mathbf{I}) = \widetilde{WB}(u\|s_i w\|\mathbf{I})$	`prop:B_si`(ii), L2733–2737	13/13
boundary shift, multiplier $\frac{1-q}{1-qw_1}$	`prop:B_shift`(ii), L2757–2760	11/11

The first of these is itself a prediction and not a transcription: I took type C's explicit diagonal and replaced *only* the boundary-root factor $(1 - w_{i_a}^{-2})$ by $(1 - w_{i_a}^{-1})$, and that matched.

Plants on V3, with admissibility decided by arithmetic. Type-C boundary root planted into the type-B diagonal: fired 10/10. Type-C shape planted into the boundary-shift multiplier: fired 11/11. Third plant, **reported in full because it did not fire everywhere**: feed the engine $W^B/\mathcal{E}^C$ — the 2026-10-17 defect itself — and compare against the type-B diagonal. It fired 8/10. The two non-firing arms are $\mathbf{M} = \mathbf{I} = [N]$ at $N = 2$ and $N = 3$, and they are **inadmissible, not blind**: by Remark 1, $\mathcal{E}^B/\mathcal{E}^C = \prod_{k\notin\mathbf{M}}(1-qw_k^2)/(1-qw_k)$, an *empty* product when $\mathbf{M} = [N]$, so there $\mathcal{E}^B = \mathcal{E}^C$ exactly and the planted object equals the true one. Verified symbolically, 28/28 for the ratio and $\mathcal{E}^B = \mathcal{E}^C$ exactly on the $\mathbf{M} = [N]$ stratum. **So the 2026-10-17 defect is invisible on the full stratum** — which is a fact about where that error could not have been caught, and it is why the plant is reported as 8/10 with a mechanism rather than as 10/10 after quietly dropping two arms.

V1 — the headline, two mechanisms. ρ^α measured at every positive root of B_2 and B_3 (nine roots: three A-type, three boundary, three pair), type-B normalisation, every $\mathbf{M}$ in the colliding stratum and every $\mathbf{I}$, against $\prod_{m\in D(\alpha)} \psi_B(w_m)^{-1}$. Mechanism M1 restricts to H_α and then divides; mechanism M2 divides as rational functions and then restricts along a uniformiser t, so that a 0/0 on H_α is resolved by cancellation in t rather than silently. All three divisors admit a *rational* parametrisation ($w_j = w_i$; $w_m = 1$; $w_b = 1/w_a$), so no radical uniformiser is introduced — the 2026-10-18 trap in which one family's order came out 2 by one script and 0 by another. **Result:** 104/104 **discriminating instances, M1 and M2 each agreeing with the prediction and with each other.** Separately, 124/124 instances of the vacuous stratum $r > |s_\alpha\mathbf{M}|$ vanish on both sides (Proposition 18), and 40/40 further instances have $\widetilde{A}_{s_\alpha\mathbf{M},\mathbf{I}} = 0$ for triangularity reasons; these 164 are reported as accounting, *not* scored as content, since $0 = \rho \cdot 0$ carries no bits about ρ.

Plants on V1. Three, each fired 13/13 on its admissible domain: ψ replaced by the type-C shape $(1-qw^2)/(1-q)$; multiplier forced to 1; ψ in place of ψ^{-1}. Each of the three is **inadmissible on nine of the thirteen root-family arms**: the three A-type and six boundary arms (across

$N = 2, 3$) have true multiplier 1 and $\psi_B(1) = 1$, so all three plants coincide with the truth there and cannot fire. Only the four pair-family arms are admissible, and on them the plants fire 13/13. Reported this way rather than as 13/13 after quietly dropping nine arms — an inadmissible plant and a blind instrument produce the same silence.

V4 — type-independence of the unnormalised ratio (Lemma 11). Two mechanisms: (S) w symbolic, reaching $|\mathbf{R}| + |\mathbf{D}| \leq 3$; (Nw) w numeric at a random point of the torus with q symbolic, reaching 6. Results: $\mathbf{D} = \{1\}$, 18/18 under (S) and 18/18 under (Nw); $\mathbf{D} = \{x, x^{-1}\}$, 7/7 under (S) and 18/18 under (Nw); $\mathbf{D} = \{x, x^{-1}, y, y^{-1}\}$, 18/18 under (Nw) only — $n = 6$ is out of reach symbolically, and I say so rather than imply two readings where there is one. The plant — prefactor *not* cancelled by line 3c, i.e. an extra $\prod_y (y + 1)$ — fired 18/18 on every case including those at -1. The runs at $\mathbf{D} \ni -1$ returned 0/18, and **that reading was wrong**: see Remark 6.

V6 — the cancelled form, and the refutation. The cancelled form of Lemma 5 agrees with the engine's literal W^B at generic u, 18/18 — which is the check that licenses using it at $u = -1$. Head-slot term at a τ-fixed point ($N = 2, 3$, $r = 1, 2$, every $\mathbf{I}$; nine instances): vanishes 9/9 at $+1$, fails to vanish 9/9 at -1 — a clean split on the nose of the stated discriminator. Adjunction law at $\mathbf{D} = \{-1\}$, symbolic in w and q, $N = 2, 3$, $|\mathbf{R}| \leq 2$, every $\mathbf{I}$: $r = 0$ **gives** 6/6 **and** $r \geq 1$ **gives 0/9** (Theorem 22). On the smaller domain $|\mathbf{R}| \leq 1$ the same phenomenon reads 4/9, the four greens again being exactly the $r = 0$ instances. Adjunction law at $\mathbf{D} = \{1\}$: 9/9; at $\mathbf{D} = \{x, x^{-1}\}$: 9/9; at $\mathbf{D} = \{1, -1\}$: 4/9, again exactly the $r = 0$ instances. One green is reported as carrying no information: $\mathbf{D} = \{-1, x, x^{-1}\}$ is 2/2, but $d = 3$ forces $|\mathbf{R}| = 0$ hence $r = 0$ on the domain reached, so it tests the law and *not* the head restriction. Every other green and every red falls on the side Lemma 7 predicts.